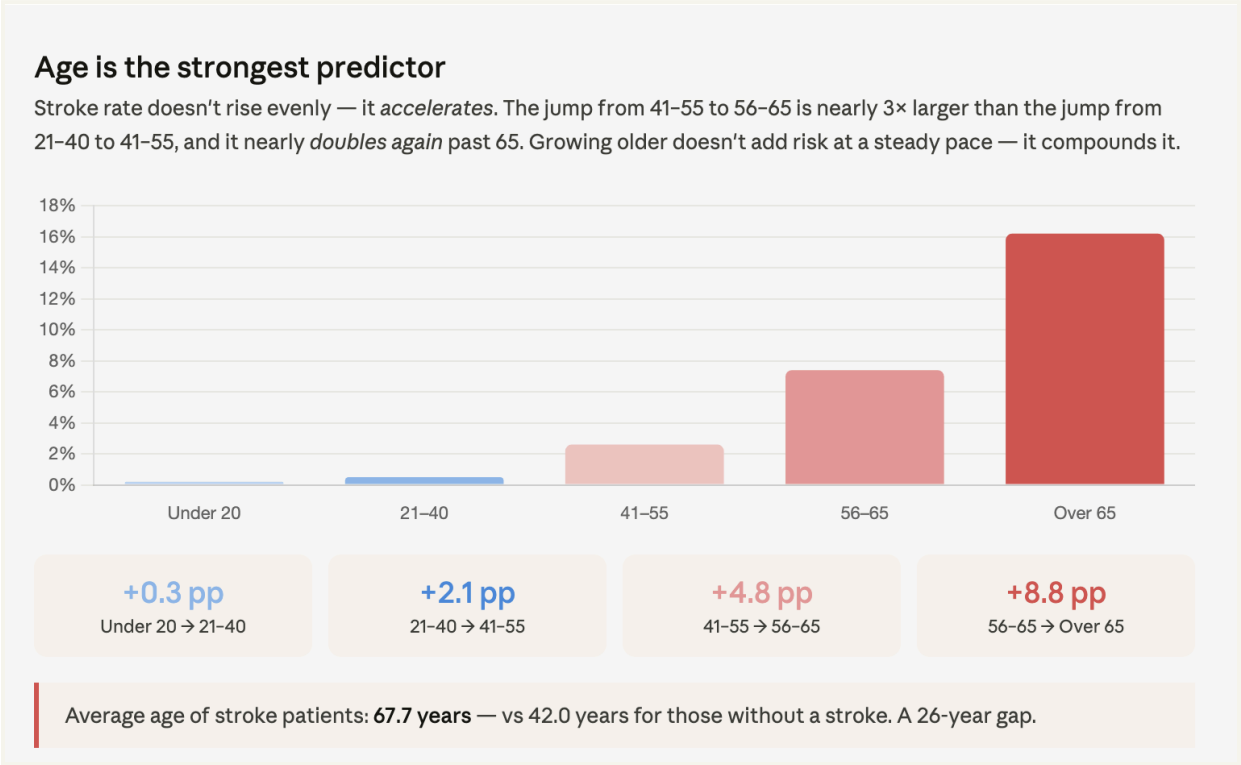
Stroke Doesn’t Happen at Random

Although stroke affects only a small portion of the population, it is rarely a matter of chance. This visualization examines health records from over 5,000 individuals to uncover the patterns and conditions most strongly linked to stroke. We find

4.9%

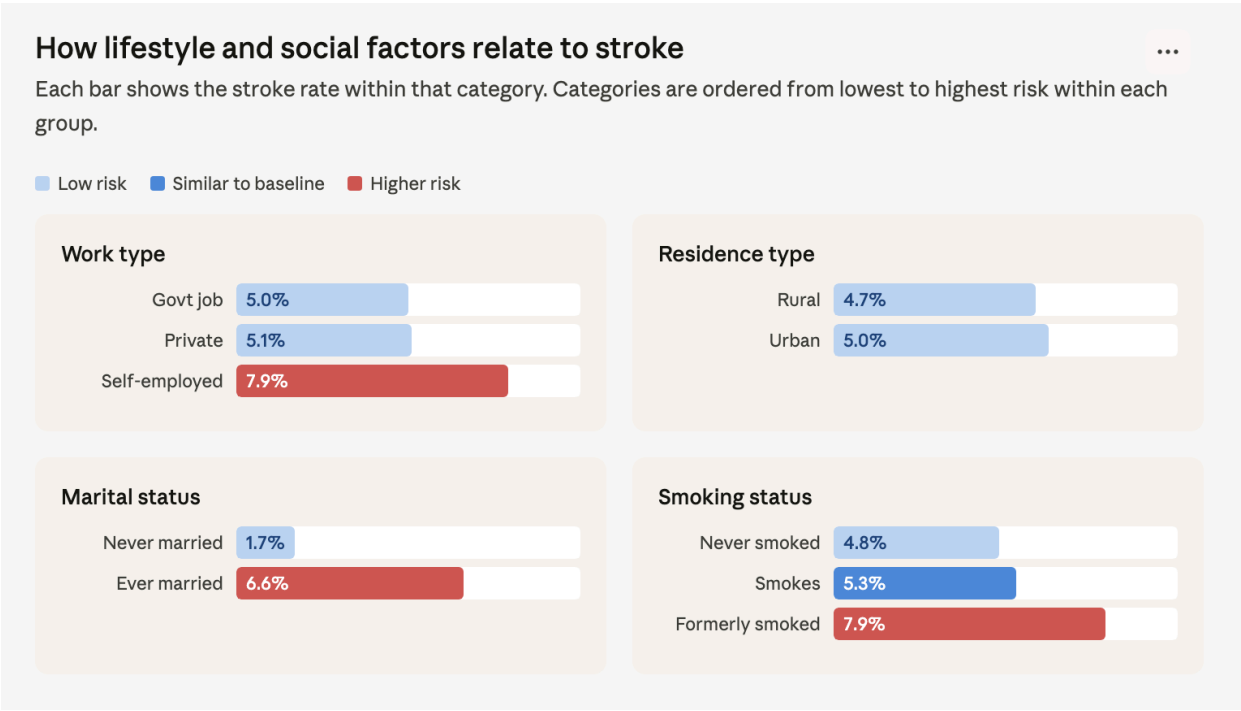
of the 5,110 patients in this dataset had a stroke

249 strokes · 4,861 no strokes



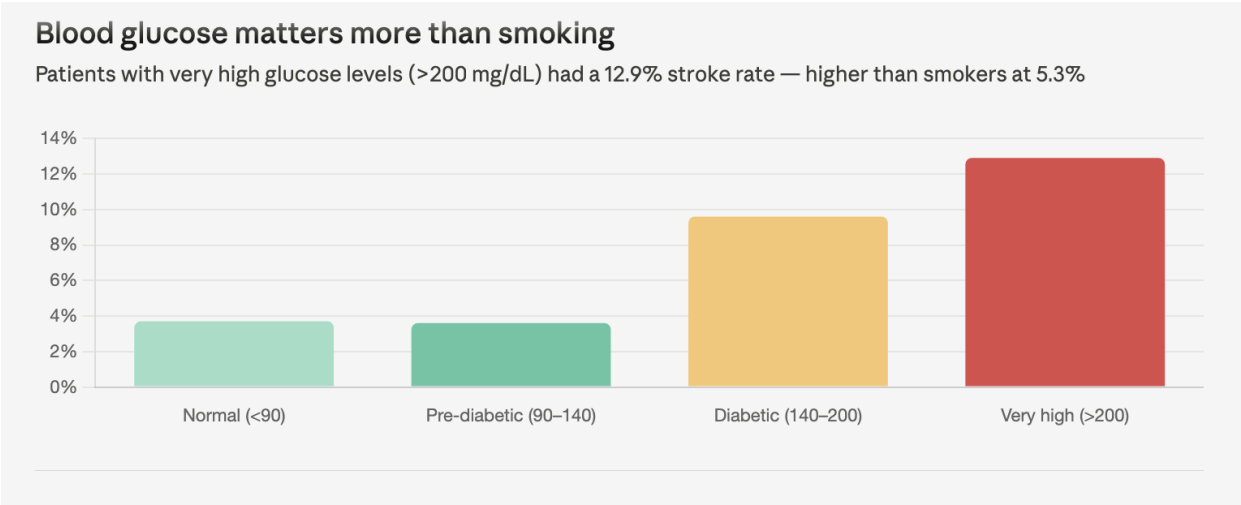
a) This chart is intended to show that stroke risk increases sharply with age, and more importantly, that the increase accelerates rather than remaining constant, highlighting age as the strongest predictor in the dataset.

b) I modified the AI's original suggestion by adding the percentage point increases between age groups, which makes the accelerating pattern more explicit and helps the audience understand not just that risk increases, but how quickly it grows at different stages of life.



a) This chart is intended to show how stroke rates vary across different lifestyle and social categories, highlighting that some groups, such as self-employed individuals and formerly smokers, are associated with noticeably higher stroke rates.

b) I modified the AI's original suggestion by reorganizing the factors into clearer, real-world categories (such as work type, residence, marital status, and smoking status), making the comparisons more intuitive for a general audience. I also adjusted the color encoding and ordering so that categories progress from lower to higher risk within each group, and used a neutral color for "smokes" because its risk level is close to the baseline, avoiding misleading emphasis.



a) This chart is intended to show how stroke risk increases across different blood glucose levels, highlighting a clear upward trend and demonstrating that very high glucose is associated with substantially higher stroke rates.
b) I kept the AI's original design unchanged because the chart already provides a clear and intuitive comparison across glucose categories, with a well-structured progression that effectively communicates the increasing risk without adding unnecessary complexity.

Surprising findings

- Smoking shows a weaker effect than expected
- Former smokers have higher stroke rates than current smokers
- Marital status appears related, but likely reflects age differences

What this data tells us

Stroke does not occur by chance. Instead, it follows clear patterns driven by age and key health conditions such as heart disease, hypertension, and high blood glucose. Although lifestyle and social factors show some variation, their effects are less consistent and often intertwined with age. One notable finding is that very high blood glucose levels are associated with a higher stroke rate than smoking in this dataset. Taken together, these results suggest that stroke risk is not evenly distributed across the population, but concentrated among individuals with specific health profiles.